

Exercise 1

Consider split conformal prediction with $\mathcal{Y} = \mathbb{R}$ and a given model $\hat{f} : \mathcal{X} \rightarrow \mathbb{R}$. Construct a score function $s(x, y)$ that would yield intervals of the form $\mathcal{C}(x) = [\hat{f}(x) - \gamma, \hat{f}(x) + 2\gamma]$ for some $\gamma > 0$.

Solution Put

$$s(x, y) = \begin{cases} 2|\hat{f}(x) - y| & y < \hat{f}(x) \\ |\hat{f}(x) - y| & y \geq \hat{f}(x). \end{cases}$$

Then, with $\hat{q}$ being a quantile of the scores on the validation set and $\mathcal{C}(x) = \{y \in \mathcal{Y} \mid s(x, y) \leq \hat{q}\}$, we will have the desired interval with $\gamma = \frac{\hat{q}}{2}$. ////